

Md. Nazmul

Department of Computer Science and Engineering, College of Technology, National University, Gazipur 1704, Bangladesh

Email: nazmulhas36@gmail.com

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To: Editor-in-Chief, IEEE Transactions on Information Forensics and Security (TIFS), IEEE Signal Processing Society
Subject: Resubmission of Manuscript T-IFS-28992-2026: “Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage”

Dear Editor-in-Chief and Editorial Board,

We are pleased to resubmit our original research manuscript titled “Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage” (Manuscript ID: T-IFS-28992-2026) for consideration in IEEE Transactions on Information Forensics and Security (TIFS).

Research Context & Methodological Innovations: Anti-Money Laundering (AML) surveillance represents a critical domain at the intersection of financial cybersecurity and machine learning. Practical surveillance is hindered by velocity evasion, topological camouflage through commercial hubs, extreme class skew ($\pi < 0.05\%$), cold-start isolation, and uncalibrated uncertainty. To resolve these interconnected bottlenecks, we propose C-STGB (Conformal Spatio-Temporal GraphBoost), featuring:

- **Tri-Band Continuous Harmonic Attention:** Disentangles microsecond smurfing bursts from 90-day dormant holding chains via continuous temporal encodings and localized Hawkes arrival intensities.
- **Context-Aware Edge Trust Gating ($\hat{g}_{ij}$):** Prunes 65.9% of spurious camouflage links to commercial hubs, mitigating feature over-smoothing.
- **Typology-Clustered Latent GraphSMOTE:** Synthesizes minority representations along empirical latent manifolds with bilinear edge scoring, recovering minority recall ($10.33\% \rightarrow 90.10\%$) with flow conservation guarantees.
- **Evidence-Adaptive Graph-Tabular Fusion:** Dynamically routes representation weight to 12-D canonical flow invariants for disconnected cold-start entities ($\deg(u) = 0$).
- **Class-Conditional Conformal Risk Control (CRC):** Provides finite-sample coverage guarantees ($1 - \alpha \geq 99.0\%$) with explicit DKW temporal drift degradation bounds, routing $> 99.4\%$ of volume through decisive tiers.
- **Multi-Agent Compliance Verification:** Compiles audit-traceable SAR dossiers structured under FinCEN Report 111 and Federal Reserve SR 26-2 model governance principles.

Empirical Evaluation & Findings: Benchmarked across 14 diverse financial transaction networks (9.53M entities, 9.53M transactions, 182 evaluation pairs over five locked seeds), C-STGB achieves an unweighted macro F1 of 67.26% (0.6848 PR-AUC; 99.44% on Elliptic-v1, 100.00% on Elliptic-v2, 96.94% on XBlock-ETH, 99.80% on PaySim Extended). It demonstrates statistically significant superiority over deep GNNs ($W = 105.0, p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ under Wilcoxon signed-rank tests) while matching tuned gradient boosted trees (CatBoost 67.14%, XGBoost 67.92%) and delivering marked gains on complex relational networks (+9.98 pp on IBM-AMLSim HI). C-STGB achieves a streaming Fast-Path inference latency of 0.45 ms per transaction event.

Declarations & Formatting Compliance: The submission package comprises our master manuscript (strictly 13.0 pages) and an accompanying Supplementary Material document formatted strictly to 5 double-column pages (10 pt font), fully complying with the IEEE Signal Processing Society ≤ 6 -page ceiling. Extended hyperparameter grids and ablation logs are mirrored in our public repository: <https://github.com/NazmulHasanNihal/Intelligent-AML>. This manuscript represents original work not published or under consideration elsewhere. All authors approved the submission and declare no competing interests.

Sincerely yours,

Md. Nazmul (Corresponding Author, nazmulhas36@gmail.com, ORCID: 0009-0001-6115-7023)

Musrat Jahan Gungun (ORCID: 0009-0006-4249-9198)

Maheli Ahmed (Faculty Supervisor, ORCID: 0000-0002-5183-7498)

Department of Computer Science and Engineering, College of Technology, National University, Bangladesh